Manual & Automation Testing | Version 1.0 | 31 October 2026

Application: https://bstackdemo.com/ | QA Team: 6 members

Document control

Field	Value
Document ID	TP-STACKDEMO-001
Application	StackDemo — https://bstackdemo.com/
Prepared by	QA Project Team (6 members)
Source	Test plan.pdf, pp. 1–10 /User Stories (US-01–US-10)
Automation	Selenium WebDriver with Java

1. Product Analysis / Overview

StackDemo is a demonstration e-commerce web application used to exercise product browsing and shopping workflows. The proposed test scope is based on ten team-authored user stories (US-01–US-10). A team member observed a username dropdown, password field, successful return to the products page, and username display after login. Remaining features and business rules must be confirmed during exploratory walkthroughs.

Target users: visitors browsing the catalog and authenticated users completing supported shopping activities. The project team will document any difference between the proposed stories and actual application behavior as a clarification item.

2. Test Objectives

- Verify the behavior of confirmed core shopping functions against reviewed acceptance criteria.
- Validate positive, negative, and boundary scenarios where supported by the application.
- Check cart accuracy, navigation, checkout validation, and order confirmation across critical end-to-end paths.

- Evaluate UI consistency and supported browser compatibility.
- Automate selected stable, high-value smoke and regression scenarios using Selenium WebDriver with Java.
- Record defects with reproducible steps and evidence, and provide execution and coverage reports.

Proposed measurable completion targets: execute or explicitly mark Blocked/Not Run for 100% of approved planned test cases; resolve or formally accept all open Critical/High defects before sign-off; report pass/fail/block rates and automated coverage against the selected automation suite. Targets require team approval.

3. Test Scope

3.1 Proposed in Scope

Requirement	Functional area	Scope decision
US-01	User Login	Proposed: verify in application
US-02	View Products	Proposed: verify in application
US-03	Search Products	Conditional: confirm search feature
US-04	Filter Products by Brand	Proposed: verify in application
US-05	Sort Products by Price	Proposed: verify in application
US-06	Add Product to Cart	Proposed: verify in application
US-07	View Shopping Cart	Proposed: verify in application
US-08	Remove Product from Cart	Proposed: verify in application
US-09	Provide Checkout Information	Proposed: verify in application
US-10	Complete Order	Proposed: verify in application

Manual coverage: functional, UI, exploratory, positive/negative, integration of visible user workflows, smoke, and regression testing. Automation coverage: selected deterministic, stable critical paths; browser selection and execution frequency to be finalized.

3.2 Out of Scope (proposed)

- Real financial transactions and external payment gateway certification.
- Direct database and backend-internal verification without access or requirements.
- Standalone API contract testing, performance/load/stress testing, and penetration testing.
- Native Android/iOS application testing.
- Features not present in the accessible demo, including search if absent.

4. Test Approach / Strategy

4.1 Manual Testing

Review acceptance criteria; design traceable test cases; prepare positive, negative and boundary data; execute smoke testing before full functional testing; log defects with severity and priority; perform retesting after fixes and targeted regression. Exploratory sessions will confirm the real UI and discover undocumented behavior.

4.2 Automation Testing

Implement UI automation with Selenium WebDriver and Java. Adopt a maintainable Page Object Model, explicit waits, independent tests, assertions, test data isolation and screenshots/logs on failure. Select a Java test runner (TestNG or JUnit) and dependency/build tool (Maven or Gradle) after team agreement. Do not assume CI integration or parallel execution until configured.

4.3 Prioritization and test design

Prioritize login, product selection, cart and checkout as critical flows. Apply equivalence partitioning, boundary value analysis where input constraints exist, decision tables for validation rules, state transition analysis for cart/session states, and exploratory testing for uncertain behavior.

4.4 Entry, Exit and Suspension Criteria

Criterion	Proposed condition
Entry	Accessible test environment; reviewed scope and test cases; test accounts and data available; browsers and automation setup ready.
Exit	Planned tests executed or dispositioned; no unaccepted Critical/High defects; defects and coverage documented; summary report reviewed by team.
Suspension	Application unavailable, authentication consistently blocked, or critical flow failure prevents meaningful testing.
Resumption	Blocking issue removed; smoke tests pass; test data and environment revalidated.

5. Test Deliverables

Deliverable	Owner / format
Reviewed user stories & acceptance criteria	Team; Word
Test plan	QA lead/editor; Word
Test scenarios and test cases	Module owners; spreadsheet or test management tool
Requirement traceability matrix (RTM)	QA team; spreadsheet
Automation code and run evidence	Automation owners; Java repository and reports
Defect reports	All testers; agreed issue tracker
Execution report and final test summary	QA lead/editor; report

6. Dependencies

- Stable access to StackDemo and supported test accounts/credentials.
- Availability of required catalog and cart data for repeatable test runs.
- Team approval of feature list and revised acceptance criteria.
- Compatible Java JDK, browser(s), Selenium library and WebDriver/browser management.
- Access to a shared repository, issue tracker and agreed communication channel.
- Clarification of any undocumented behavior, particularly product search and checkout validation.

7. Test Environment

Component	Planned configuration
AUT	https://bstackdemo.com/ (public demo environment)
OS	Team workstations; record actual OS/version per execution
Browsers	Google Chrome baseline; Edge/Firefox if team resources permit, versions recorded
Manual tools	Browser Developer Tools, test case repository, screenshots
Automation	Selenium WebDriver + Java; runner/build tools TBD
Source control	Git/GitHub (proposed)
Defect tracking	Jira or agreed tracker (TBD)
Test data	Demo usernames and authorized test passwords; no real customer data

8. Risk Management

Risk	Impact	Mitigation
Demo environment unavailable or reset	High	Run early smoke check; preserve evidence; reschedule

		blocked tests.
Requirements differ from live UI	High	Perform feature walkthrough; log clarifications; update scope and RTM.
Unstable UI selectors / timing	Medium	Use robust locators, explicit waits, Page Objects; avoid brittle sleeps.
Shared cart/account state between testers	Medium	Allocate accounts or reset state; coordinate parallel runs.
Time constraints across six members	High	Prioritize critical flows; time-box exploratory sessions; track daily progress.
Unsupported browser or driver versions	Medium	Pin tested versions; validate setup before automation work.
Search feature absent	Medium	Mark US-03 not applicable pending review; remove related cases from baseline.

9. Schedule and Milestones

Illustrative 10-working-day schedule. Replace day labels with the instructor's actual deadline and team capacity before approval.

Days	Activity	Output
1–2	Product walkthrough, requirements review, test planning	Confirmed scope; approved draft plan
3–4	Test scenario/case design, RTM, environment setup	Reviewed cases and smoke checklist
5–6	Manual functional execution and defect logging	Execution records and defects

5–8	Automation development and debugging (parallel)	Selected automated regression suite
7–9	Retesting, regression, browser checks	Updated execution and defect status
10	Final review and reporting	Test summary and project handover

10. Roles and Responsibilities

Proposed module-based assignment; adjust after workload estimation. All six members contribute to manual test design/execution and selected automation. One member coordinates plan updates and reporting; a second reviewer validates deliverables.

Role	Primary module / duty
Member 1 / Team Leader	Login; planning and status consolidation
Member 2	Catalog and product display
Member 3	Search (if available), brand filters, sorting
Member 4	Add/remove cart and cart contents
Member 5	Checkout information and validation
Member 6	Order completion and end-to-end regression

Communication: daily brief team check-in; record blockers and defects in a shared tracker; review requirements changes before updating test cases; peer-review test cases and automation pull requests.

11. Defect Management and Reporting

Every defect should contain an ID, concise title, environment, preconditions, reproducible steps, expected and actual results, severity, priority, evidence, status and owner. Proposed lifecycle: New _ Triaged _ In Progress _ Ready for Retest _ Closed/Reopened; use the team tracker's actual workflow. Distinguish requirement gaps from reproducible defects.

12. Traceability and Metrics

Map US-01–US-10 and each approved AC to manual test case IDs and applicable automated test IDs. Report cases planned/executed/passed/failed/blocked, defects by severity and status, requirements covered, and automation pass/fail counts. Do not claim execution results until tests run.

13. Assumptions, Open Items and Approval

- US-03 product search is unconfirmed; confirm before finalizing in-scope features.
- Exact checkout fields, validation messages, cart total rules, and order status behavior are unconfirmed.
- Login dropdown, password field, product-page redirect and username display are based on a team member's observed interaction.
- Agree test runner, build tool, target browsers, issue tracker, repository, named owners and actual dates.
- Approval: QA team / instructor (if required) pending.

References / Basis

Course material: "Test plan (1).pdf", especially pp. 1–2 (components), pp. 5–7 (planning process, entry/exit criteria, deliverables and schedule), pp. 8–10 (sample and updates). Team-provided draft US-01–US-10 and reported login observations. No official StackDemo product specification was provided.